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Chemical Composition and Nutritional and Medicinal Value of Fruit Bodies and Submerged Cultured Mycelia of Culinary-Medicinal Higher Basidiomycetes Mushrooms

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ABSTRACT: This research gives the results of a proximate analysis (moisture, ash, crude protein, fat, total carbohydrates, and total energy); a bioactive compounds analysis (γ-aminobutyric acid [GABA], ergothioneine, lovastatin, and cordycepin); fatty acid and amino acid analysis; and an analysis of macro- and microelement content of fruit bodies and mycelia of 15 higher Basidiomycetes medicinal mushroom strains belonging to 12 species. The results obtained demonstrate that almost all investigated mushrooms were found to be good sources of proteins and carbohydrates, with content varying in the ranges of 8.6–42.5% and 42.9–83.6%, respectively. Different species exhibited distinct free amino acid profiles. The total amino acid content was highest in *Ophiocordyceps sinensis* (MB) (23.84 mg/g) and *Cordyceps militaris* (FB) (23.69 mg/g). The quantification of the identified fatty acids indicated that, in general, palmitic acid, oleic acid, stearic acid, and linoleic acid were the major fatty acids. The micro- and macroelement compositions were studied, and the highest results were (as milligrams per kilogram) 224–7307 for calcium, 1668–38564 for potassium, 1091–11676 for phosphorus, and 5–97 for zinc. Bioactive components were lovastatin, GABA, and ergothioneine, which are commonly found in most mushrooms. *C. militaris* (FB), *Pleurotus ostreatus* (FB), and *Coprinus comatus* (FB) were most abundant and contained a high amount of GABA (756.30 µg/g, 1304.99 µg/g, 1092.45 µg/g, respectively) and ergothioneine (409.88 µg/g, 2443.53 µg/g, 764.35 µg/g, respectively). The highest lovastatin content was observed in *Hericium erinaceus* (FB) (14.38 µg/g) and *Ganoderma lucidum* (FB) (11.54 µg/g). In contrast to *C. militaris* (FB), cordycepin was not detected in *O. sinensis* (MB). The fruit body biomass of *C. militaris* cordycepin content reached 1.743 mg/g dry weight. The nutritional values of the mushroom species studied here could potentially be used in well-balanced diets and as sources of bioactive compounds.

KEY WORDS: medicinal mushrooms, fruiting body, mycelium, proximate composition, amino acids, fatty acids, macroelements, microelements, lovastatin, γ-aminobuthiric acid (GABA), ergothioneine, cordycepin

ABBREVIATIONS: FB, fruit body; **GABA,** γ-aminobuthiric acid; **HPTLC,** high-performance thin layer chromatography; **MB,** mycelial biomass.

I. INTRODUCTION

Many higher Basidiomycetes mushroom species have been used as a source of food and pharmaceutical compounds. In many parts of the world, higher Basidiomycetes mushrooms are one of the most popular foods, not only because of their texture and flavor but also their chemical and medicinal characteristics. Many mushrooms, from a pharmaceutical point of view, contain biologically active substanc-

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es in fruit bodies, cultured mycelia, and cultured broth.[1–3]

A higher Basidiomycetes mushrooms possess certain natural advantages in terms of dietary supremacy over the rest of the vegetarian diet.These include high protein content, all essential amino acids (thus able to act as a substitute for meat), a chitinous wall that acts as a source of dietary fiber, high vitamin content (B_1, B_2, B_{12}, C, D, and E), micro- and macroelements, carbohydrates, low fat content, and virtually no cholesterol.[2,4–6] Higher Basidiomycetes mushrooms could also be a source of many different nutritive and medicinal components such as unsaturated fatty acids, phenolic compounds, tocopherols, ascorbic acid, and carotenoids, which are particularly important for the regulation of physiological functions in the human organism. Therefore, these mushrooms could be introduced directly into the diet to promote health and take advantage of the additive and synergistic effects of all the bioactive compounds present.[7–10] A total of 126 medicinal functions are thought to be produced by medicinal mushrooms, including antitumor, anti–human immunodeficiency virus, immunomodulating, antioxidant, radical scavenging, cardiovascular, antihypercholesterolemia, antiviral, antibacterial, antiparasitic, antifungal, detoxification, hepatoprotective, antidiabetic, and many other effects.[1,2,11–16]

In relation to the use of mushrooms as a functional food and as sources for the development of drugs and nutraceuticals, we are interested in the evaluation of the chemical and nutritional composition of some widely consumed medicinal mushroom species and hence in assessing their pharmaceutical potential.

Several higher Basidiomycetes mushrooms, Because of their capability for producing an array of bioactive metabolites such as lovastatin, γ-aminobutyric acid (GABA), and ergothioneine constitute an important source of novel drugs. Lovastatin (mevinolin) is one of the best-known hypocholesterolemic pharmacological agents of natural origin. It is the competitive inhibitor of 3-hydroxy-3-methylglutaril-coenzyme A reductase, the key enzyme in cholesterol metabolism. Second to reducing cholesterol production, lovastatin regulates the low-density lipoprotein receptor activity, and as a result cholesterol concentration decreases.[17] Using normal clinical doses (80 mg/day), lovastatin reduces blood cholesterol. In a clinical study including 16 elderly patients between 70 and 78 years of age who received lovastatin (Mevacor) 80 mg/day, the mean inhibitory activity of 3-hydroxy-3-methylglutaril-coenzyme A reductase in plasma increased approximately 45% compared with 18 patients between 18 and 30 years of age.[18]

GABA is one of the major inhibitory neurotransmitters in the central nervous system and suppresses peripheral sympathetic neurotransmission. An important role for GABA in both central and peripheral nervous system control of blood pressure has been demonstrated.[19] It also has an immune-amplifying role in neuroinflammatory diseases such as multiple sclerosis.[20]

L-ergothioneine (a water-soluble thiol) is a histidine derivative with sulfur on the imidazole ring and a trimethylated amine.[21,22] Numerous *in vitro* assays demonstrated the antioxidant and cytoprotective capabilities of ergothioneine against a wide range of cellular stressors.[23–25] Ergothioneine has been shown to be radioprotective and to scavenge singlet oxygen, hydroxyl radicals, hypochlorous acid, and peroxyl radicals as well as to inhibit peroxynitrite-dependent nitration of proteins and DNA.[26] However, ergothioneine is apparently only produced by fungi and some prokaryotes.[27,28] Ergothioneine can only be absorbed from the diet and cannot be biosynthesized in humans.

Medicinal species from genera *Ophiocordyceps* (*O. sinensis*) and *Cordyceps* (*C. militaris*) produce various bioactive compounds, among which cordycepin (3-deoxyadenosine) is a major bioactive component. Cordycepin displays broad-spectrum biological activities including antifungal, antitumor, and antiviral activity.[29–31] Cordycepin also has been shown to regulate the production of interleukins in T lymphocytes.[32]

The primary goal of this research was to examine the chemical composition and nutritional value of fruiting bodies and mycelia of 15 higher Basidiomycetes medicinal mushroom strains be-

longing to 12 species: *Agaricus brasiliensis*, *Coprinus comatus*, *C. militaris*, *Flammulina velutipes*, *Ganoderma lucidum*, *Grifola frondosa*, *Hericium erinaceus*, *Lentinus edodes*, *O. sinensis*, *Pleurotus ostreatus*, *Trametes versicolor*, and *Tremella fuciformis*. These mushrooms have been recommended by the United Nations Food and Agriculture Organization as highly nutritious edible mushrooms and as sources of valuable medicinal compounds.

II. MATERALS AND METHODS

A. Mushroom Materials

Fifteen higher Basidiomycetes medicinal mushroom strains belonging to 12 species used in this study were *A. brasiliensis* (mycelium and fruit body [FB]), *C. comatus* (FB), *C. militaris* (FB), *F. velutipes* (FB), *G. lucidum* (mycelium and FB), *G. frondosa* (FB), *H. erinaceus* (mycelium and FB), *L. edodes* (FB), *O. sinensis* (mycelium), *P. ostreatus* (FB), *T. versicolor* (mycelium), and *T. fuciformis* (FB). All mushroom strains were obtained from the culture collection of Mycolivia Ltd. and maintained on malt extract agar plates at 5°C.

B. Inoculum Preparation and Cultivation Conditions

The inocula were prepared by growing mushroom species on a rotary shaker at 150 rpm in 250-mL flasks containing 100 mL of synthetic medium: glucose, 15 g L^{-1}; peptone, 3.0 g L^{-1}; KH_2PO_4, 0.8 g L^{-1}; K_2HPO_4, 0.2 g L^{-1}; $MgSO_4 \cdot 7H_2O$, 0.5 g L^{-1}; and yeast extract, 5.0 g L^{-1}. Initial pH of the media was 6.0. After 5–7 days of cultivation, mycelial pellets were harvested and homogenized with a Warning laboratory blender. To accumulate a high yield of fungal biomass, the higher Basidiomycetes underwent submerged cultivation at 27°C on a rotary shaker (150 rpm) in 2-L Erlenmeyer flasks containing 1 L of synthetic medium. The basal medium consisted of glucose (20 g L^{-1}) and peptone (5.0 g L^{-1}), and nutrient composition was slightly changed according to the mushroom used.[33,34] After 7–18 days of submerged cultivation, mycelial biomasses (MBs) were harvested by filtration.

The mushroom fruit bodies were grown by Mycolivia Ltd. (Israel) according to the methods described by Stamets.[35] Mushroom mycelia and fruit bodies were dried at 50°C to a constant weight and milled to powder for extraction.

C. Proximate Chemical Analysis

The proximate chemical composition of mushroom samples—moisture, ash, crude fat, and crude protein—was determined according to the methods described by the Association of Official Analytical Chemists.[36] Total carbohydrate content was calculated by subtracting the contents of ash, fat, and protein from 100 and is expressed as a percentage of dry mass. Total reducing sugars were determined using the 3,5-dinitrosalicylic acid method.[37,38] The energy provided by 100 g of dry mushrooms was calculated as follows:

Energy (kcal/100 g) =
TC × 4 + Fat × 9 + Protein × 4,

where TC = total carbohydrates.

D. Fatty Acid Analysis

Fatty acids were extracted from the dried samples using the method described by Hamilton and Hamilton.[39] Fatty acids were converted into their methyl esters and analyzed by gas chromatography using an HP model 5890A–5971A gas chromatograph (Hewlett-Packard, Germany) equipped with 100 m × 0.25 mm Fused Silica Capillary Column (SP-2560, Supelco Ltd.) with a 0.20-μm film thickness. The sample was injected using HP 7673 GC/SFC auto injector model 18593B (Agilent Co.). The temperature program was 140°C for 5 minutes, increased to 240°C at 4°C/minute, and maintained at 240°C for 15 minutes. Helium (flow rate, 20 cm/s) was used as the carrier gas. The temperature for the injection port and the flame ionization detector was 260°C. Fatty acid methyl esters were identified by comparing their retention times to an authentic standard mixture (Supelco 37 Component FAME

Mix; Sigma, St. Louis, MO, USA).

E. Amino Acid Analysis

Powdered FBs and MBs were hydrolyzed with 6 N hydrogen chloride for 24 hours at 105°C, and the amino acids in the hydrolysates were derivatized using a Water PICO-TAG workstation. The derivatized amino acids were analyzed by liquid chromatography consisting of Waters 515 pumps, a Waters 486 detector, and a Reodyne injector equipped with a Waters PICO-TAG column (3.9 × 150 mm). Amino acids were identified by comparing their retention times to those of an authentic standard mixture.[40]

F. Macro- and Microelement Analysis

For determination of macro- and microelements, powdered FBs and MBs were analyzed by inductively coupled plasma optical emission spectroscopy (ICAP 600 series; Thermo Electron Corp.). Solutions of each element were prepared from analytical-grade chemicals (Merck) using high-purity water.

G. Determining Bioactive Compounds

1. Lovastatin, GABA, and Ergothioneine

Lovastatin, GABA, and ergothioneine were extracted and analyzed according to methods described previously[41,42] and quantified by the calibration curve of the respective authentic standard.

2. Cordycepin

Cordycepin in *C. militaris* (FB) and *O. sinensis* (MB) was determined by high-performance thin layer chromatography (HPTLC) using a method reported elsewhere.[43] Cordycepin standard was from Sigma-Aldrich Inc. Chloroform, methanol, and water were analytical grade. Ethanol was distilled before use.

Powdered *C. militaris* (FB) (0.5 g) and *O. sinensis* (MB) (10.8 g) were extracted separately with hot chloroform/methanol (2:1, v/v) and ethanol using Soxhlet extraction for 3 hours. Powdered FBs (2 g) and MBs (2 g) also were extracted with water at ambient temperature for 24 hours. The liquid extracts were membrane-filtered and concentrated under reduced pressure. The residue was dissolved in 5 mL methanol and used to quantify cordycepin.

The extracts were analyzed using a CAMAG HPTLC system. Chromatography was performed on silica gel 60 F_{254} TLC precoated aluminum plates (20 × 10 cm, layer thickness, 0.2 mm; Merck) and used as a stationary phase. A stock solution (1 mg/mL) of cordycepin was prepared by dissolving 10 mg cordycepin standard, accurately weighed, in 5 mL water in a 10-mL volumetric flask and diluting to volume with water. Standard solutions containing 200, 250, 300, 350, 400, 450, and 500 μg/mL were prepared by separately diluting 2, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0 mL of the stock solution, respectively, with 10 mL water. The standard (1.5–4.5 μL) and samples (10–15 μL) were applied to the plates as 6-mm bands (slit dimension, 5.00 × 0.45 mm), applied at 8 mm, starting 15 mm from the edge of the plate, by means of a Linomat 5 applicator equipped with 100-μL syringe. The plates were developed to a distance of 80 mm in a twin trough chamber previously saturated with mobile-phase chloroform:methanol (3:1, v/v) for 20 minutes. After removal from the chamber, the plates were completely dried in air at ambient temperature, scanned, and quantified at 263 nm in absorbance/reflectance mode using the CAMAG TLC scanner 3 and winCATS 3.2.1 software incorporating the track optimizing option. The standard plot of cordycepin was established by plotting the peak area versus concentration (nanograms per microliter) corresponding to each spot.

H. Statistical Analysis

All the analyses were performed in triplicate, and the results were averaged. The data were subjected to analysis of variance for a completely randomized design using SAS software (SAS Institute, Cary, NC, USA).

III. RESULTS AND DISCUSSION

A. Proximate Chemical Analysis

The results of the proximate chemical composition (expressed on a dry weight basis) for the investigated mushroom species are shown in Table 1. Crude protein, moisture, ash, fats, carbohydrates, and energy value varied according to the species. Considerable differences in proximate chemical composition of FBs and MBs also were observed.

The major chemical components of mushrooms are proteins and carbohydrates. It was reported that the protein contents of mushrooms are affected by a number of factors, namely, the type of mushroom, the stage of development, the part sampled, and the amount of nitrogen available.[44] In this study, protein content in all the tested mushroom species varied from 8.6% to 42.5%. The highest protein content was found for *A. brasiliensis* FB (37.3%), *H. erinaceus* MB (42.5%), *O. sinensis* MB (38.6%), and *P. ostreatus* FB (33.5 %). Protein content determined in *P. ostreatus* FB and *O. sinensis* MB were higher than those reported previously for these species (24.9% and 14.8–30.4%, respectively).[45,46] The lowest protein content was found for *T. versicolor* MB (8.6%), *G. lucidum* FB (9.20%), and *T. fuciformis* FB (13.0%). *T. versicolor* and *T. fuciformis* have been reported to contain low protein content (4.2% and 8.3%, respectively).[38]

In general, carbohydrate contents of mushrooms have been reported to be in the range of 33.5–72.2%.[5,10,47] In this study, the total carbohydrate content of *T. versicolor* MB, *G. lucidum* FB, and *T. fuciformis* FB were highest (83.5%, 83.6%, and 72.9%, respectively) and were comparable to those reported previously for these species (88.3%, 85.2%, and 84.6%, respectively).[38] The lowest carbohydrate content was assayed in *O. sinensis* MB (44.8%) and *P. ostreatus* FB (48.9%) and was much lower than those reported (61.2% in *O. sinensis*, 66.4% in *P. ostreatus*, and 85.8–88.4% in genus *Pleurotus*) by other researchers.[38,47]

The fat content determined in all the mushroom species examined varied from 0.8% to 7.9%. In *O. sinensis* MB, *H. erinaceus* MB, *G. frondosa* FB, and *H. erinaceus* MB the fat content was relatively higher (7.9%, 6.3%, 5.3%, and 5.1%, respectively) than rest of the mushroom species, in which the fat content was between 0.8 and 2.9%.

The ash content of the mushrooms is usually between 5% and 12% of dry weight.[6,48,49] In our work, the ash content varied among species from 1.0% to– 13.2%. *A. brasiliensis* FB, *H. erinaceus* FB, and *C. comatus* FB showed higher ash content (8.9%, 6.8%, and 13.2%, respectively) than the other investigated mushrooms. The lowest amounts of ash were observed in MB (2.7%) and FB (1.0%) of *G. lucidum* and in *T. versicolor* MB (3.1%).

The dry weight of fresh mushrooms is generally 5–15%, and the nutritional profile of mushrooms is directly influenced by moisture content. This means that moisture content is between 85% and 95%. Moreover, the dry mass of a biological material is not absolutely free from moisture and generally contains about 8–10% moisture. The moisture content of all investigated mushrooms in this study was significantly different and varied from 2.8% to 8.2%.

Total energy values of the mushroom species examined were varied from 337 to 405 kcal/100 g. Thus, the investigated mushrooms were high in crude protein (except *T. versicolor* MB, *G. lucidum* FB, and *T. fuciformis* FB) and carbohydrates and low in fat and ash content. Similar findings have been reported previously.[5,47]

B. Amino Acid Contents

Results of the analysis of the mushroom species for essential and nonessential amino acids are shown in Table 2, which reveals the presence of 17 amino acids. Amino acid composition of mushrooms are reported to be comparable to animal proteins.[50] Total amino acid content varied from 2.80 to 30.65 mg/g. Similar findings have previously been published.[16,51] The essential amino acids detected in all the mushrooms were threonine, valine, methionine, isoleucine, leucine, phenylalanine, lysine, histidine, and arginine (threonine, valine, and leucine are the ma-

TABLE 1: Proximate Chemical Composition of Higher Basidiomycetes Mushrooms on Dry Weight Basis

Species	Portion of Mushroom	Protein (%)	Moisture (%)	Ash (%)	Fat (%)	Total Carbohydrates (%)	Energy (kcal/100 g)
Agaricus brasiliensis	FB	37.3 ± ± 0.22	6.5 ± ± 0.11	8.9 ± 0.09	2.4 ± 0.03	44.9 ± 2.5	350 ± 12
	MB	22.3 ± 0.31	4.9 ± 0.09	8.6 ± 0.16	2.2 ± 0.03	62.0 ± 6.3	357 ± 16
Coprinus comatus	FB	22.7 ± 0.37	4.2 ± 0.06	13.2 ± 0.42	1.3 ± 0.02	58.6 ± 5.1	337 ± 11
Cordyceps militaris	FB	29.7 ± 0.42	7.7 ± 0.61	5.4 ± 0.16	2.9 ± 0.18	54.3 ± 5.5	362 ± 19
Flammulina velutipes	FB	23.4 ± 0.19	5.0 ± 0.13	8.3 ± 0.08	2.1 ± 0.10	61.2 ± 4.3	357 ± 16
Ganoderma lucidum	FB	9.20 ± 0.32	5.1 ± 0.16	1.0 ± 0.00	1.1 ± 0.01	83.6 ± 4.4	381 ± 14
	MB	25.2 ± 0.24	5.7 ± 0.12	2.7 ± 0.02	2.8 ± 0.04	63.6 ± 3.4	380 ± 23
Grifola frondosa	FB	18.3 ± 0.34	4.8 ± 0.08	4.7 ± 0.07	5.3 ± 0.09	66.9 ± 8.4	389 ± 13
Hericium erinaceus	FB	**20.8** ± 0.43	**6.2** ± 0.14	**6.8** ± 0.22	**5.1** ± 0.11	**61.1** ± 3.6	374 ± 14
	MB	42.5 ± 0.18	3.9 ± 0.08	4.4 ± 0.32	6.3 ± 0.32	42.9 ± 2.8	398 ± 18
Lentinus edodes	FB	18.5 ± 0.16	7.3 ± 0.10	5.1 ± 0.05	0.8 ± 0.01	68.3 ± 4.7	354 ± 18
Ophiocordyceps sinensis	MB	38.6 ± 0.53	2.8 ± 0.06	5.9 ± 0.14	7.9 ± 0.25	44.8 ± 5.3	405 ± 22
Pleurotus ostreatus	FB	33.5 ± 0.22	8.2 ± 0.07	7.1 ± 0.06	2.3 ± 0.07	48.9 ± 2.7	350 ± 20
Trametes versicolor	MB	8.60 ± 0.51	2.8 ± 0.06	3.1 ± 0.11	2.0 ± 0.12	83.5 ± 6.9	386 ± 17
Tremella fuciformis	FB	13.0 ± 0.12	5.5 ± 0.18	6.5 ± 0.14	2.1 ± 0.08	72.9 ± 6.4	363 ± 13

FB, fruiting body; MB, mycelial biomass.

jor ones), whereas 8 nonessential amino acids were present, including cysteic acid, aspartic acid, serine, glutamic acid, proline, glycine, alanine, and tyrosine; these were dominated by glutamic acid and aspartic acid. Total amino acids in FBs and MBs were found to be significantly different. For example, *G. lucidum* MB accumulated 15.14 mg/g of total amino acids, which was 5.4-fold higher than in the FB (2.8 mg/g). However, in the MB of *H. erinaceus* the amount of amino acids was 30.65 mg/g, 2-fold higher than in the FBs (14.33 mg/g). *F. velutives* FB produced considerably higher amounts of nonessential amino acids (14.98 mg/g), which was almost 2 times higher than the content of essential amino acids (7.88 mg/g). Our results indicate the importance of combining different species of mushrooms in the human diet to obtain both a healthy amount and a variety of all essential amino acids.

C. Fatty Acid Content

Results for fatty acid composition, shown in Table 3, reveal the presence of 18 fatty acids (8 saturated and 10 unsaturated fatty acids) in the mushroom species analyzed. These varied in quantity, and a significant difference in the fatty acid profile determined for the FBs and MBs also was observed. All the mushroom species contained a good amount of essential fatty acids (linoleic acid [C18:2n6] and linolenic acid [C18:3n6]) except *A. brasiliensis* MB. Among the saturated fatty acids, palmitic acid (C16:0) was the major one, amounting to 13.65 mg/g (*A. brasiliensis* FB) to 27.5 mg/g (*A. brasiliensis* MB), whereas the unsaturated fatty acid fraction was dominated by oleic acid (C18:1n9) and linoleic acid, the contents of which ranged from 3.46 mg/g (*L. edodes* FB) to 52.4 mg /g (*G. lucidum* FB) and 3.60 mg/g (*A. brasiliensis* MB) to 75.8 mg /g (*L. edodes* FB), respectively. *G. lucidum* FB, *G. frondosa* FB, *H. erinaceus* MB, *H. erinaceus* FB, and *A. brasiliensis* MB seemed to be good sources of oleic acid and produced 52.4, 44.1, 37.2, 33.7, and 52.4 mg/g, respectively. Linoleic acid was present in significant quantities in *A. brasiliensis* FB (65.86 mg/g), *L. edodes* FB (75.8 mg/g), C. *militaris* FB (58.7 mg/g), *G. lucidum* MB (42.4 mg/g), *C. comatus* FB (71.1 mg/g), and *F. velutipes* FB (51.2 mg/g). Stearic acid (C18:0), the amounts of which varied from 1.88 mg/g (*L. edodes* FB) to 18.9 mg/g (*T. versicolor* MB), also was found in all the tested mushrooms. The other fatty acids detected were found only in small or trace amounts (<6.6%) or not detected. These findings are in agreement with results reported by several other authors.[52–54]

D. Macro- and Microelement Contents

We detected 29 micro- and macroelements. The average contents, expressed as milligrams per kilograms (dry weight of mushrooms), are given in Table 4. Minerals and micronutrient concentrations varied by fungal species. In addition, the comparison of FB and MB showed significant differences in results. For example, the potassium concentration ranged from 2538 mg/kg in *T. versicolor* MB to 43351 mg/kg in *C. comatus* FB; sulfur from 598 mg/kg in *G. lucidum* FB to 6848 mg/kg in *L. edodes* FB; iron from 39 mg/kg in *L. edodes* FB to 18125 mg/kg in *A. brasiliensis* MB; and magnesium from 576 mg/kg in *G. lucidum* FB to 3460 mg/kg in *A. brasiliensis* MB. *A. brasiliensis* FB and *O. sinensis* MB demonstrated the same and highest content of phosphorus (11676 mg/kg); the lowest concentration (1091 mg/kg) was determined in *G. lucidum* FB. Potassium, sulfur, iron, magnesium, and phosphorus have been reported to be the most abundant elements encountered in various species of edible mushrooms.[55,56] The most toxic metals studied were cadmium (Cd), lead, mercury, copper, arsenic, silver , and nickel. Cd is known as a principle toxic element, since it inhibits many life processes by exerting toxic effects on several types of cells, tissues, and organs. Other pathologies induced by this metal include atherosclerosis, cardiovascular diseases, and rapid bone demineralization and osteoporosis.[57–59] FBs and MBs of *G. lucidum* and *H. erinaceus* contained the lowest amount of Cd (0.1 mg/kg). It was shown that Cd content in *Agaricus* spp. ranged between 0.5 and 5.0 mg/kg. In the present study, the highest

TABLE 2: Content of Amino Acids of Higher Basidiomycetes Mushrooms on Dry Weight Basis

Amino Acids	Mushroom Species/Content (mg/g dry weight)														
	AB (FB)	*AB* (MB)	*LE* (FB)	*OS* (MB)	*CM* (FB)	*GF* (FB)	*TV* (MB)	*PO* (FB)	*GL* (MB)	*GL* (FB)	*HE* (FB)	*HE* (MB)	*CC* (FB)	*FV* (FB)	*TF* (FB)
Cystic acid	0.28 ± 0.02	0.43 ± 0.04	0.35 ± 0.06	0.33 ± 0.02	0.55 ± 0.03	0.21 ± 0.06	0.14 ± 0.02	0.33 ± 0.02	0.41 ± 0.01	0.12 ± 0.00	0.32 ± 0.07	0.62 ± 0.04	0.27 ± 0.06	0.39 ± 0.08	0.24 ± 0.03
Aspartic acid	1.74 ± 0.05	1.93 ± 0.02	0.97 ± 0.08	2.51 ± 0.32	2.87 ± 0.42	0.91 ± 0.08	0.56 ± 0.04	2.15 ± 0.24	1.47 ± 0.18	0.30 ± 0.06	1.30 ± 0.06	3.88 ± 0.43	1.67 ± 0.22	2.50 ± 0.26	0.98 ± 0.01
MS	0.38 ± 0.01	0.43 ± 0.05	0.28 ± 0.02	0.53 ± 0.08	0.58 ± 0.11	0.19 ± 0.03	0.11 ± 0.00	0.51 ± 0.11	0.53 ± 0.06	0.09 ± 0.01	0.53 ± 0.02	0.77 ± 0.04	0.31 ± 0.10	0.55 ± 0.04	0.15 ± 0.04
Threonine	1.16 ± 0.06	1.06 ± 0.08	0.70 ± 0.08	1.57 ± 0.22	1.91 ± 0.32	0.69 ± 0.05	0.36 ± 0.02	1.43 ± 0.07	0.97 ± 0.04	0.22 ± 0.05	1.18 ± 0.11	2.04 ± 0.13	1.17 ± 0.08	1.01 ± 0.11	0.34 ± 0.03
Serine	0.88 ± 0.04	0.96 ± 0.06	0.54 ± 0.06	1.28 ± ± 0.11	1.22 ± 0.10	0.54 ± 0.04	0.26 ± 0.01	1.02 ± 0.08	0.70 ± 0.02	0.16 ± 0.03	0.83 ± 0.06	1.66 ± 0.09	0.71 ± 0.04	1.44 ± 0.06	0.59 ± 0.07
GA	2.54 ± 0.11	2.99 ± 0.11	1.43 ± 0.13	2.51 ± 0.24	3.00 ± 0.66	1.09 ± 0.14	0.45 ± 0.04	2.88 ± 0.35	2.92 ± 0.36	0.28 ± 0.06	1.36 ± 0.09	4.80 ± 0.28	2.32 ± 0.36	4.42 ± 0.39	1.11 ± 0.10
Proline	1.14 ± 0.05	1.35 ± 0.07	0.38 ± 0.03	1.45 ± 0.03	0.87 ± 0.07	0.21 ± 0.03	0.15 ± 0.03	0.83 ± 0.05	1.08 ± 0.08	0.15 ± 0.04	0.51 ± 0.03	1.36 ± 0.06	0.69 ± 0.03	1.02 ± 0.07	0.44 ± 0.06
Glycine	0.87 ± 0.03	0.99 ± 0.05	0.43 ± 0.05	1.65 ± 0.14	1.06 ± 0.05	0.47 ± 0.07	0.26 ± 0.06	0.94 ± 0.07	0.80 ± 0.04	0.19 ± 0.02	0.75 ± 0.05	1.63 ± 0.08	0.70 ± 0.02	1.17 ± 0.05	0.52 ± 0.02
Alanine	1.29 ± 0.11	1.09 ± 0.03	0.56 ± 0.07	1.78 ± 0.09	1.39 ± 0.12	0.56 ± 0.06	0.29 ± 0.08	1.18 ± 0.12	0.78 ± 0.02	0.18 ± 0.03	0.89 ± 0.07	1.56 ± 0.12	0.91 ± 0.08	1.61 ± 0.08	0.60 ± 0.04
Valine	2.58 ± 0.14	1.04 ± 0.08	1.69 ± 0.21	1.82 ± 0.16	0.96 ± 0.13	1.52 ± 0.22	0.32 ± 0.03	1.13 ± 0.08	0.95 ± 0.06	0.19 ± 0.07	0.90 ± 0.04	1.61 ± 0.04	0.81 ± 0.06	1.07 ± 0.02	0.41 ± 0.02

TABLE 2: Content of Amino Acids of Higher Basidiomycetes Mushrooms on Dry Weight Basis (continued)

Amino Acids	Mushroom Species/Content (mg/g dry weight)														
	AB (FB)	*AB* (MB)	*LE* (FB)	*OS* (MB)	*CM* (FB)	*GF* (FB)	*TV* (MB)	*PO* (FB)	*GL* (MB)	*GL* (FB)	*HE* (FB)	*HE* (MB)	*CC* (FB)	*FV* (FB)	*TF* (FB)
Isoleucine	0.78 ± 0.10	0.91 ± 0.08	0.35 ± 0.04	1.23 ± 0.07	1.42 ± 0.08	0.48 ± 0.11	0.32 ± 0.05	0.83 ± 0.02	0.73 ± 0.03	0.24 ± 0.05	0.77 ± 0.01	1.61 ± 0.02	0.75 ± 0.04	0.98 ± 0.06	0.34 ± 0.04
Leucine	1.26 ± 0.02	1.84 ± 0.06	0.55 ± 0.06	1.83 ± 0.33	1.79 ± 0.42	0.65 ± 0.13	0.33 ± 0.02	1.27 ± 0.06	1.12 ± 0.24	0.26 ± 0.05	0.85 ± 0.06	2.69 ± 0.18	0.98 ± 0.05	1.78 ± 0.16	0.66 ± 0.04
Tyrosine	0.58 ± 0.05	0.79 ± 0.12	0.25 ± 0.04	0.85 ± 0.05	1.66 ± 0.07	0.32 ± 0.08	0.12 ± 0.01	0.68 ± 0.03	0.47 ± 0.02	0.07 ± 0.02	0.46 ± 0.03	1.36 ± 0.06	0.76 ± 0.02	0.94 ± 0.09	0.48 ± 0.07
Phen	0.69 ± 0.01	1.04 ± 0.04	0.37 ± 0.06	1.13 ± 0.08	0.88 ± 0.02	0.41 ± 0.05	0.20 ± 0.03	0.79 ± 0.08	0.71 ± 0.08	0.13 ± 0.04	0.58 ± 0.03	1.64 ± 0.05	0.60 ± 0.06	0.99 ± 0.07	0.41 ± 0.06
Lysine	1.08 ± 0.01	0.41 ± 0.00	0.48 ± 0.03	1.38 ± 0.14	1.40 ± 0.19	0.56 ± 0.07	0.22 ± 0.03	1.27 ± 0.12	0.45 ± 0.04	0.08 ± 0.01	0.97 ± 0.08	0.93 ± 0.10	0.77 ± 0.08	1.05 ± 0.03	0.46 ± 0.02
Histidine	0.41 ± 0.03	0.53 ± 0.06	0.22 ± 0.01	0.51 ± 0.05	0.42 ± 0.01	0.29 ± 0.02	0.10 ± 0.00	0.44 ± 0.03	0.28 ± 0.01	0.04 ± 0.00	0.53 ± 0.05	0.63 ± 0.04	0.43 ± 0.05	0.45 ± 0.04	0.15 ± 0.00
Arginine	1.10 ± 0.13	0.75 ± 0.09	0.52 ± 0.05	1.47 ± 0.12	1.72 ± 0.42	0.82 ± 0.10	0.23 ± 0.04	1.64 ± 0.01	0.77 ± 0.03	0.10 ± 0.02	1.07 ± 0.07	1.86 ± 0.15	0.97 ± 0.11	1.49 ± 0.20	0.60 ± 0.06
Total	18.76 ± 0.11	18.54 ± 0.08	10.09 ± 0.6	23.84 ± 0.13	23.69 ± 0.19	9.92 ± 0.08	4.42 ± 0.03	19.32 ± 0.18	15.14 ± 0.08	2.8 ± 0.03	14.33 ± 0.12	30.65 ± 0.22	14.82 ± 0.17	22.84 ± 0.11	8.48 ± 0.09

MS, methionine sulfon; GA, glutamic acid; Phen, phenylalanine; *AB*, *Agaricus brasiliensis*; *CC*, *Coprinus comatus*; *CM*, *Cordyceps militaris*; *FV*, *Flammulina velutipes*; FB, fruiting body; *GF*, *Grifola frondosa*; *GL*, *Ganoderma lucidum*; *HE*, *Hericium erinaceus*; *LE*, *Lentinus edodes*; MB, mycelial biomass; *OS*, *Ophiocordyceps sinensis*; *PO*, *Pleurotus ostreatus*; *TF*, *Tremella fuciformis*; *TV*, *Trametes versicolor*.

TABLE 3: Content of Fatty Acids of Higher Basidiomycetes Mushrooms on Dry Weight Basis

	Mushroom Species/Content (mg/g dry weight)														
Fatty Acids	***AB* (FB)**	***AB* (MB)**	***LE* (FB)**	***OS* (MB)**	***CM* (FB)**	***GF* (FB)**	***TV* (MB)**	***PO* (FB)**	***GL* (MB)**	***GL* (FB)**	***HE* (FB)**	***HE* (MB)**	***CC* (FB)**	***FV* (FB)**	***TF* (FB)**
C14:0	0.26 ± 0.02	0.40 ± 0.03	0.35 ± 0.06	0.30 ± 0.08	0.20 ± 0.03	ND	1.10 ± 0.07	0.60 ± 0.10	ND	0.40 ± 0.04	0.30 ± 0.03	ND	0.40 ± 0.02	0.8 ± 0.03	ND
C15:0	0.56 ± 0.07	ND	2.00 ± 0.15	ND	0.70 ± 0.06	0.56 ± 0.06	1.60 ± 0.14	2.60 ± 0.28	ND	0.80 ± 0.04	1.30 ± 0.06	ND	0.50 ± 0.04	1.7	ND
C16:0	13.65 ± 0.26	27.5 ± 0.37	13.7 ± 0.33	20.7 ± 1.62	18.0 ± 1.11	16.85 ± 2.41	**27.2 ± 2.68**	25.8 ± 3.52	15.2 ± 2.05	20.1 ± 1.83	24.0 ± 1.68	14.0 ± 3.67	17.5 ± 3.02	15.9 ± 2.04	**19.0 ± 3.01**
C16:1	ND	1.10 ± 0.06	0.60 ± 0.04	1.00 ± 0.04	1.50 ± 0.06	0.50 ± 0.09	1.20 ± 0.08	ND	0.30 ± 0.04	0.60 ± 0.05	0.50 ± 0.05	ND	0.50 ± 0.04	0.80 ± 0.10	**0.40 ± 0.02**
C17:0	0.55 ± 0.04	0.30 ± 0.04	ND	ND	0.70 ± 0.02	ND	ND	ND	ND	0.30 ± 0.07	ND	ND	ND	ND	ND
C17:1	0.60 ± 0.01	0.30 ± 0.07	ND	ND	0.60 ± 0.04	ND	ND	4.70 ± 1.13	ND	0.60 ± 0.03	ND	ND	ND	ND	ND
C18:0	6.20 ± 0.11	12.2 ± 0.61	1.88 ± 0.25	5.20 ± 0.41	4.60 ± 0.22	2.34 ± 0.62	18.9 ± 3.11	17.7 ± 2.24	3.70 ± 0.62	7.80 ± 0.86	8.10 ± 0.22	4.1 ± 0.37	2.00 ± 0.10	2.80 ± 0.21	**4.90 ± 0.44**
C18:1(n-7)	0.56 ± 0.04	1.30 ± 0.12	ND	ND	2.20 ± 0.08	ND	0.80 ± 0.06	ND	ND	ND	4.70 ± 0.08	0.8 ± 0.71	0.70 ± 0.62	1.00 ± 0.02	**1.80 ± 0.06**
C18:1(n-9)	5.75 ± 0.09	45.7 ± 3.24	3.46 ± 0.31	36.3 ± 3.51	11.1 ± 1.06	44.1 ± 5.21	16.2 ± 1.62	17.1 ± 1.87	29.3 ± 4.08	52.4 ± 5.55	33.7 ± 0.48	37.2 ± 3.41	5.70 ± 1.01	10.7 ± 1.04	**51.8 ± 4.44**
C18:2(n-6)	65.86 ± 2.32	3.60 ± 0.22	75.8 ± 4.33	32.9 ± 3.16	58.7 ± 3.72	35.1 ± 4.34	13.7 ± 1.47	14.7 ± 2.06	42.4 ± 5.31	11.6 ± 1.04	26.9 ± 4.92	32.8 ± 2.89	71.1 ± 6.41	51.2 ± 4.24	**17.2 ± 2.13**
C18:3(n-3)	1.10 ± 0.07	0.30 ± 0.03	0.28 ± 0.04	2.10 ± 0.09	0.30 ± 0.05	ND	1.70 ± 0.18	6.60 ± 0.98	2.20 ± 0.52	1.70 ± 0.73	0.40 ± 0.32	1.00 ± 0.02	1.10 ± 0.07	13.0 ± 2.02	**0.70 ± 0.10**

TABLE 3: Content of Fatty Acids of Higher Basidiomycetes Mushrooms on Dry Weight Basis (continued)

	Mushroom Species/Content (mg/g dry weight)														
Fatty Acids	***AB* (FB)**	***AB* (MB)**	***LE* (FB)**	***OS* (MB)**	***CM* (FB)**	***GF* (FB)**	***TV* (MB)**	***PO* (FB)**	***GL* (MB)**	***GL* (FB)**	***HE* (FB)**	***HE* (MB)**	***CC* (FB)**	***FV* (FB)**	***TF* (FB)**
C18:3(n-6)	0.36 ± 0.05	0.20 ± 0.07	1.05 ± 0.02	ND	1.40 ± 0.07	0.55 ± 0.06	4.30 ± 0.23	3.20 ± 0.67	ND	1.80 ± 0.32	ND	ND	ND	ND	ND
C20:0	1.10 ± 0.06	0.90 ± 0.05	ND	0.26 ± 0.02	ND	ND	ND	0.70 ± 0.06	1.00 ± 0.04	ND	ND	1.50 ± 0.10	ND	ND	**0.60** ± 0.04
C20:1	ND	0.30 ± 0.02	ND	ND	ND	ND	ND	1.10 ± 0.02	1.10 ± 0.07	ND	ND	1.40 ± 0.06	ND	ND	ND
C22:0	2.60 ± 0.06	0.80 ± 0.05	ND	0.21 ± 0.01	ND	ND	2.00 ± 0.16	ND	2.70 ± 0.42	0.60 ± 0.05	ND	4.30 ± 0.02	ND	0.30 ± 0.01	0.80 ± 0.03
C22:4(n-6c)	ND	ND	ND	0.7 ± 0.04	ND	ND	ND	ND	ND	0.60 ± 0.02	ND	0.50 ± 0.02	ND	ND	ND
C24:0	0.85 ± 0.03	4.60 ± 0.14	0.88 ± 0.11	0.33 ± 0.02	ND	ND	1.40 ± 0.08	1.10 ± 0.09	1.80 ± 0.04	0.70 ± 0.06	ND	1.90 ± 0.38	0.40 ± 0.06	0.40 ± 0.03	**2.50 ±** 0.07
C24:1	ND	0.60 ± 0.06	ND	ND	ND	ND	3.40 ± 0.26	1.90 ± 0.05	ND	ND	ND	0.20 ± 0.06	ND	0.90 ± 0.05	ND

AB, *Agaricus brasiliensis*; C14:0, myristic acid; C15:0, pentadecanoic acid; C16:0, palmitic acid; C16:1, palmitoleic acid; C17:0, heptadecanoic acid; C17:1, cis-10-heptadecenoic acid; C18:0, stearic acid; C18:1(n-7), trans-vaccenic acid (omega-7); C18:1(n-9), oleic acid (omega-9); C18:2n-6, linoleic acid; C18:3n-3, linolenic acid; C18:3n-6, linolenic acid; C20:0, arachidic acid; C20:1, eicosenoic acid; C22:0, behenic acid; C22:4n-6c, docosatetraenoic acid; C24:0, lignoceric acid; C24:1, nervonic acid; *CC*, *Coprinus comatus*; *CM*, *Cordyceps militaris*; FB, fruiting body; *FV*, *Flammulina velutipes*; *GF*, *Grifola frondosa*; *GL*, *Ganoderma lucidum*; *HE*, *Hericium erinaceus*; *LE*, *Lentinus edodes*; MB, mycelial biomass ND, not detected; *OS*, *Ophiocordyceps sinensis*; *PO*, *Pleurotus ostreatus*; *TF*, *Tremella fuciformis*; *TV*, *Trametes versicolor*.

TABLE 4: Content of Micro- and Macroelements of Higher Basidiomycetes Mushrooms on Dry Weight Basis

Micro- and Macro-elements	Mushroom Species/Content (mg/kg dry weight)														
	AB (FB)	*AB* (MB)	*LE* (FB)	*OS* (MB)	*CM* (FB)	*GF* (FB)	*TV* (MB)	*PO* (FB)	*GL* (MB)	*GL* (FB)	*HE* (FB)	*HE* (MB)	*CC* (FB)	*FV* (FB)	*TF* (FB)
Aluminum	418 ± 14.08	170 ± 11.7	16 ± 1.21	45 ± 3.32	10 ± 0.84	71 ± 3.12	838 ± 18.6	53 ± 5.42	107 ± 4.44	18 ± 1.52	84 ± 5.64	62 ± 6.73	1960 ± 31.2	**58 ± 6.32**	69 ± 2.55
Arsenic	2 ± 0.11	<0.2	0.7 ± 0.04	1.0 ± 0.13	<0.2	0.7 ± 0.04	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	0.2 ± 0.01	**0.2 ± 0.04**	<0.2 ±
Barium	2 ± 0.72	65 ± 3.31	1.0 ± 0.07	3.0 ± 0.72	1 ± 0.03	2.0 ± 0.11	17 ± 2.06	1 ± 0.04	4 ± 0.63	6 ± 1.02	1 ± 0.07	11 ± 0.88	9 ± 0.65	3.0 ± 0.52	2 ± 0.32
Beryllium	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Boron	6 ± 0.36	74 ± 3.56	8.0 ± 1.21	8.0 ± 1.04	<5	19 ± 2.21	5 ± 067	<5	<5	<5	9 ± 1.21	9 ± 1.21	≤5	<5	<5 ±
Cadmium	4 ± 0.43	0.2 ± 0.04	0.8 ± 0.12	<0.1	0.1 ± 0.01	0.3 ± 0.01	0.3 ± 0.04	0.1 ± 0.02	0.1 ± 0.02	0.1 ± 0.03	0.1 ± 0.02	0.1 ± 0.03	0.2 ± 0.02	0.4 ± 0.06	0.1 ± 0.02
Calcium	474 ± 16.72	7307 ± 22.4	436 ± 18.7	2403 ± 36.5	224 ± 14.4	483 ± 12.1	2633 ± 22.4	184 ± 13.6	1301 ± 22.4	1179 ± 24.1	395 ± 2.71	5526 ± 34.5	1290 ± 17.9	467 ± 15.2	365 ± 14.7
Chromium	13 ± 1.11	6 ± 0.81	0.4 ± 0.03	2.0 ± 0.41	1 ± 0.22	2.0 ± 0.13	102 ± 13.2	1 ± 0.02	1 ± 0.04	4 ± 0.21	1 ± 0.02	1 ± 0.06	12 ± 1.10	5 ± 0.81	3 ± 0.13
Cobalt	0.4 ± 0.06	1 ± 0.06	<0.2	<0.2	<0.2	<0.2	1 ± 0.11	<0.2	<0.2	<0.2	<0.2	<0.2	0.3 ± 0.00	<0.2	<0.2
Copper	45 ± 2.54	4 ± 0.66	6.0 ± 0.81	10 ± 1.24	14 ± 2.21	24 ± 2.14	15 ± 1.04	15 ± 2.21	5 ± 1.03	4 ± 0.07	13 ± 1.15	8 ± 1.02	22 ± 3.06	1 ± 0.12	6 ± 0.54
Iron	189 ± 13.2	18125 ± 26.8	39 ± 2.14	206 ± 8.65	68 ± 4.62	112 ± 4.62	1097 ± 37.4	148 ± 12.7	115 ± 3.59	125 ± 11.3	270 ± 17.6	228 ± 13.4	772 ± 15.4	133 ± 13.5	150 ± 13.8
Lead	1 ± 0.03	4 ± 0.34	0.2 ± 0.02	1.0 ± 0.06	<0.2	0.3 ± 0.03	2 ± 0.04	0.2 ± 0.02	0.2 ± 0.04	0.3 ± 0.02	0.4 ± 0.04	0.3 ± 0.04	2 ± 0.30	0.5 ± 0.13	0.3 ± 0.01
Lithium	0.7 ± 0.04	0.2 ± 0.02	0.2 ± 0.04	<0.2	0.2 ± 0.03	0.2 ± 0.03	0.4 ± 0.02	<0.2	<0.2	<0.2	<0.2	<0.2	0.3 ± 0.04	<0.2	<0.2
Magnesium	1193 ± 22.7	3460 ± 51.1	920 ± 12.6	1804 ± 14.3	1098 ± 21.3	994 ± 6.94	689 ± 16.3	1462 ± 11.4	1717 ± 18.8	576 ± 10.4	1166 ± 16.2	2092 ± 15.1	1205 ± 11.3	748 ± 12.4	1571 ± 16.2
Manganese	15 ± 1.67	72 ± 2.51	13 ± 1.34	21 ± 2.06	5 ± 0.42	14 ± 1.42	82 ± 5.71	8 ± 0.68	55 ± 3.41	65 ± 6.41	11 ± 1.10	15 ± 1.42	39 ± 2.42	6 ± 0.92	7 ± 1.04

TABLE 4: Content of Micro- and Macroelements of Higher Basidiomycetes Mushrooms on Dry Weight Basis (continued)

Micro- and Macro-elements	Mushroom Species/Content (mg/kg dry weight)														
	AB (FB)	*AB* (MB)	*LE* (FB)	*OS* (MB)	*CM* (FB)	*GF* (FB)	*TV* (MB)	*PO* (FB)	*GL* (MB)	*GL* (FB)	*HE* (FB)	*HE* (MB)	*CC* (FB)	*FV* (FB)	*TF* (FB)
Mercury	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5
Molybdenum	0.2 ± 0.01	1 ± 0.12	<0.2	0.2 ± 0.02	<0.2	<0.2	1 ± 0.12	<0.2	1 ± 0.03	<0.2	<0.2	2 ± 0.38	0.3 ± 0.02	<0.2	<0.2
Nickel	5 ± 1.01	3 ± 0.33	<0.2	5.0 ± 1.00	0.4 ± 0.04	0.9 ± 0.07	48 ± 3.54	0.5 ± 0.02	1 ± 0.11	2 ± 0.36	<0.2	1 ± 0.02	5 ± 0.49	3 ± 0.31	1 ± 0.06
Phosphorus	11676 ± 57.7	10871 ± 41.2	5354 ± 37.6	11676 ± 51.3	8739 ± 11.5	5914 ± 16.5	1351 ± 13.6	11271 ± 15.4	4651 ± 33.0	1091 ± 16.2	6121 ± 8.73	5216 ± 18.3	9276 ± 22.1	5965 ± 18.0	9055 ± 14.2
Potassium	34623 ± 78.8	8460 ± 28.2	21379 ± 42.2	18815 ± 16.7	19641 ± 36.4	20039 ± 37.8	2538 ± 26.5	36420 ± 52.1	4812 ± 16.5	1668 ± 13.4	29163 ± 34.5	6621 ± 22.7	43351 ± 28.8	29758 ± 25.1	38564 ± 54.1
Selenium	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Silver	1 ± 0.06	<0.2	<0.2	<0.2	<0.2	0.2 ± 0.01	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Sodium	86 ± 3.82	743 ± 20.0	114 ± 1.04	86 ± 2.07	943 ± 18.1	98 ± 2.62	110 ± 8.55	216 ± 10.0	73 ± 2.37	12 ± 2.04	157 ± 13.4	116 ± 3.58	1160 ± 16.7	327 ± 14.2	366 ± 6.71
Sulfur	5254 ± 43.6	2184 ± 31.3	6848 ± 48.3	5254 ± 31.5	4306 ± 17.2	1781 ± 14.2	659 ± 25.7	3520 ± 17.7	2255 ± 18.4	598 ± 14.7	1932 ± 4.25	3285 ± 16.4	2331 ± 15.2	1406 ± 21.4	2090 ± 18.5
Strontium	1.0 ± 0.06	7 ± 0.06	1.0 ± 0.02	7.0 ± 1.12	1 ± 0.02	4.0 ± 0.67	13 ± 1.52	1 ± 0.08	3 ± 0.22	3 ± 0.13	1 ± 0.11	9 ± 0.82	4 ± 0.06	2 ± 0.04	2 ± 0.23
Tin	0.5 ± 0.02	0.4 ± 0.04	≤0.5	0.5 ± 0.04	0.5 ± 0.04	0.5 ± 0.01	1 ± 0.03	1 ± 0.05	1 ± 0.04	1 ± 0.11	1 ± 0.02	1 ± 0.06	0.4 ± 0.02	<0.5	<0.5
Titanium	2.0 ± 0.03	1 ± 0.01	0.3 ± 0.02	0.3 ± 0.03	0.2 ± 0.02	0.9 ± 0.12	12 ± 1.21	0.2 ± 0.06	0.3 ± 0.01	0.4 ± 0.10	1 ± 0.07	1 ± 0.01	5 ± 1.24	1 ± 0.06	1 ± 0.02
Vanadium	0.5 ± 0.05	<0.2	<0.2	<0.2	<0.2	<0.2	1 ± 0.07	<0.2	<0.2	<0.2	<0.2	<0.2	1 ± 0.05	<0.2	<0.2
Zinc	97 ± 6.82	24 ± 1.12	46 ± 3.33	36 ± 3.11	79 ± 2.71	74 ± 4.23	19 ± 1.24	85 ± 5.82	21 ± 2.17	5 ± 0.43	59 ± 0.46	38 ± 32.12	44 ± 5.62	48 ± 2.48	63 ± 6.51

AB, *Agaricus brasiliensis*; *CC*, *Coprinus comatus*; *CM*, *Cordyceps militaris*; FB, fruiting Bodies; *FV*, *Flammulina velutipes*; *GF*, *Grifola frondosa*; *GL*, *Ganoderma lucidum*; *HE*, *Hericium erinaceus*; *LE*, *Lentinus edodes*; MB, mycelial Biomass *OS*, *Ophiocordyceps sinensis*; *PO*, *Pleurotus ostreatus*; *TF*, *Tremella fuciformis*; *TV*, *Trametes versicolor*.

level of Cd was found only in *A. brasiliensis* FB (4.0 mg/kg).[60] In other tested species, Cd content varied between 0.1 and 0.8 mg/kg dry matter. Cd is reported to be the most deleterious heavy metal in mushrooms.[61] Lead concentrations ranged from 0.2 to 4.0 mg/kg, and the highest amount was observed in *A. brasiliensis* MB. Despite this, the Cd and lead content in most of the tested mushroom species did not exceed the statutory limits.[62]

Negative effects of aluminum on the nervous system have been reported. Human exposure to aluminum can lead to apparent age-related neurological deficits resembling Alzheimer disease in human beings.[63,64] However, few studies focused on the potential immunological responses induced by aluminum.[65] The estimated daily dietary exposure to aluminum in the general population, assessed in several European countries, varied from a mean of 0.2 to 1.5 mg/kg and up to 2.3 mg/kg in highly exposed consumers.[66] In our study a considerably high concentration of aluminum was accumulated in *C. comatus* FB (1960 mg/kg) and *T. versicolor* MB (838 mg/kg). Aluminum content in all other samples ranged from 10 to 418 mg/kg. Mercury was found only in trace amounts (<0.5 mg/kg) in all investigated mushroom species. *A. brasiliensis* MB and *T. versicolor* MB seemed to be the best source of iron and possessed the highest concentrations: 18125 and 1097 mg/kg, respectively. In all other samples significant amounts of iron were determined and ranged from 39.0 to 772.0 mg/kg. These values are in line with those reported for other mushrooms.[67,68] IN general, the highest element concentrations were found in the following mushroom species: *A. brasiliensis*, *T. versicolor*, *L. edodes*, *H. erinaceus*, and *C. comatus*.

E. Bioactive Compounds Contents

1. GABA, Ergothioneine, and Lovastatin Contents

Bioactive compounds such as GABA, ergothioneine, and lovastatin were quantified in 15 mushroom strains belonging to 12 species; the results are given in Table 5. GABA was present in all mushroom samples and varied from 13.24 to 1304.99 µg/g dry weight. The FBs of *C. militaris*, *P. ostreatus*, and *C. comatus* contained the highest amounts of GABA (756.30, 1304.99, and 1092.45 µg/g, respectively), whereas mycelia of *G. lucidum* contained the highest amount of GABA (92.71 µg/g). GABA has been shown to be absent in FBs of *G. lucidum*,[69] but in our study low GABA content (13.24 µg/g) was found in the FBs of *G. lucidum*. Compared to published results,[40] GABA content was found to be higher in FBs and MBs.

Ergothioneine was detected in all the mushroom species except *A. brasiliensis* MB and *G. lucidum* FB; the concentrations varied from 19.49 to 2443.53 µg/g. FBs of *H. erinaceus*, *P. ostreatus*, *Coprinus comatus*, and *Cordyceps militaris* contained the highest concentration of ergothioneine (629.96, 2443.53, 764.35, and 409.88 µg/g, respectively). Considerably high amounts of ergothioneine have been reported in the FBs of the genus *Pleurotus*[41]; the FBs of *P. ostreatus* seem to be an source of abundant ergothioneine. Ergothioneine has been reported in FBs of *A. brasiliensis* (84.77 µg/g) and *G. lucidum* (14.52 µg/g),[70] but in our study no ergothioneine was detected in these mushrooms.

Among all studied mushrooms, only 7 species contained lovastatin; amounts varied from 2.76 (*C. militaris*) to 14.38 µg/g (*H. erinaceus*). Lovastatin was found in the FBs of 5 species, namely, *A. brasiliensis*, *Cordyceps militaris*, *G. lucidum*, *H. erinaceus*, *Coprinus comatus*, and the mycelia of *A. brasiliensis* and *T. versicolor*. Absence of lovastatin in FBs of *F. velutipes*, *G. lucidum*, *G. frondosa*, and *L. edodes* was in agreement with published results.[41,70]

2. Cordycepin Content

The extracts of the FB of *C. militaris* and MB of *O. sinensis* using chloroform:methanol (2:1), ethanol, and water were isolated and analyzed for cordycepin. Results of the analysis are presented in Table 6. Cordycepin was found in all the 3 extracts (7.25, 8.37, and 5.28 mg/g, respectively) isolated from *C. militaris* FB. Ethanol was determined to

TABLE 5: Contents of γ-aminobutyric Acid (GABA), Ergothioneine, and Lovastatin in Higher Basidiomycetes Mushrooms on Dry Weight Basis.

	Content (μg/g dry weight)		
Species	**GABA**	**Ergothioneine**	**Lovastatin**
Agaricus brasiliensis (FB)	394.26 ± 26.24	37.36 ± 1.50	9.64 ± 0.69
Agaricus brasiliensis(MB)	38.84 ± 1.73	ND	3.61 ± 0.04
Coprinus comatus (FB)	1092.45 ± 39.74	764.35 ± 9.12	9.61 ± 0.83
Cordyceps militaris (FB)	756.30 ± 41.47	409.88 ± 27.86	2.76 ± 0.14
Flammulina velutipes (FB)	360.41 ± 3.2	98.61 ± 3.99	ND
Ganoderma lucidum (FB)	13.24 ± 0.58	ND	11.54 ± 0.73
Ganoderma lucidum (MB)	92.71 ± 4.68	219.59 ± 10.82	ND
Grifola frondosa (FB)	37.52 ± 0.66	207.00 ± 13.69	ND
Hericium erinaceus (FB)	42.93 ± 2.17	629.96 ± 36.80	14.38 ± 0.99
Hericium erinaceus (MB)	56.00 ± 0.84	149.24 ± 5.51	ND
Lentinus edodes (FB)	186.17 ± 2.95	334.01 ± 10.01	ND
Ophiocordyceps sinensis (MB)	81.40 ± 0.29	52.18 ± 2.75	ND
Pleurotus ostreatus (FB)	1304.99 ± 40.26	2443.53 ± 135.18	ND
Trametes versicolor (MB)	48.72 ± 1.13	119.70 ± 2.22	6.99 ± 0.24
Tremella fuciformis (FB)	524.88 ± 3.61	19.49 ± 0.53	ND

ND-Not detected; FB – Fruiting Bodies; MB – Mycelial Biomass.

be the most efficient solvent for the high-yield extraction of cordycepin. The concentration of cordycepin in the fungal biomass of *C. militaris* was determined to be 1.743 mg/g, which is significantly high. This is in agreement with published reports wherein the capability of *C. militaris* to synthesize high amounts of cordycepin was demonstrated.[71,72] Thus, the FB of *C. militaris* could be considered a potential source for the production of cordycepin. Occurrence of cordycepin in both natural and cultivated *O. sinensis* has been reported.[42] In our study, however, cordycepin was not detected in any of the 3 extracts (chloroform:methanol, ethanol, and water) isolated from the *O. sinensis* MB.

IV. CONCLUSIONS

Knowledge of the full chemical composition, bioactive compounds, and nutritional values of medicinal mushroom species was limited until the past 20 years. In recent years, much attention has been devoted to the study of the digestibility and bioavailability of nutrients from medicinal mushrooms, and it has dynamically expanded.

In the frame of this research, the proximate composition and content of amino acids, fatty acids, micro- and macroelements, and bioactive compounds were studied in 15 higher Basidiomycetes medicinal mushroom strains belonging to 12 species from different taxonomic groups, which are the most popular and regularly consumed in the world. Information about the nutritional values of some of the investigated mushroom species is limited in the literature.

In general, all studied mushrooms, especially *A. brasiliensis*, *G. frondosa*, *P. ostreatus*, *T. versicolor*, *H. erinaceus*, *C. comatus*, and *C. militaris*, contained useful nutraceuticals such as GABA, ergothioneine, lovastatin, cordycepin, essential amino acids, and fatty acids, which are well known for their benefits in human health. The results show that the analyzed mushroom species possess mod-

TABLE 6: Cordycepin Content in Cordyceps militaris (FB) and *Ophiocordyceps sinensis* (MB)

Species	Chloroform / methanol extract (2:1 v/v)	Ethanol extract	Water extract	Biomass
Cordyceps millitaris (FB)	7.25 mg/g	8.37 mg/g	5.28 mg/g	1.743 mg/g
Ophiocordyceps sinensis (MB)	ND	ND	ND	ND

ND-Not detected; FB – Fruiting Bodies; MB – Mycelial Biomass.

erate amino acid and fatty acid contents, which may be relevant from a nutritional point of view because these compounds are indispensable for human health. From these results, it can be seen that all investigated medicinal mushrooms in this work can be considered as rich and suitable sources of nutrients. At the same time, they demonstrated low contents of toxic metals (lead, Cd, arsenic, mercury, and silver) and large amounts of calcium, iron, potassium, and phosphorus, which are favorable for human health. Thus, the combination of rich nutritional compositions in these mushroom FBs and MBs makes them appropriate as functional foods and nutritional supplements.

Our results showed that most mushroom FBs and mycelia, especially FBs of *P. ostreatus*, *C. comatus*, and *C. militaris*, contained substantial amounts of ergothioneine and GABA. Lovastatin was not detected in all samples. Only FBs of *A. brasiliensis*, *G. lucidum*, *H. erinaceus*, and *C. comatus* had considerably high amounts of lovastatin. They remain an important source of novel drugs because of their cholesterol-lowering, blood pressure–controlling, and antioxidant properties.

Various mushrooms, such as *O. sinensis* and *C. militaris*, have been extensively used as crude drugs. Both of these species contain many kinds of active components; the major active constituent is cordycepin (3-deoxyadenosine) and, because of its various physiological activities, it is now used for multiple medicinal purposes. From the 1960s until now, many biological functions of cordycepin have been discovered. Cordycepin has antitumor, antimetastatic, antibacterial, antiviral, immunomodulatory, and anti-inflammatory activities. In the past 20 years, studies targeting cordycepin as a therapeutic agent have been in progress. In our study, MB of *O. sinensis* did not show cordycepin content, but the FB of *C. militaris* produced a significant amount. It is necessary to mention that cordycepin production reported by a number of researchers was much lower than what was revealed in this study. Therefore, because of the ability of *C. militaris* to produce much higher amounts of cordycepin than *O. sinensis*, it would be the most efficient way to meet the increasing demand for cordycepin produced by *C. militaris*.

The ultimate goal of this research is to describe the nutritional components and bioactive compounds of medicinal mushrooms for their pharmacological use. With regard to the quantitative profile of the different compounds studied, the different mushroom species were very distinct. It is therefore important to combine different mushroom species in the human diet to obtain optimum amounts of nutraceutical compounds and a great diversity of dietary supplements.

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